

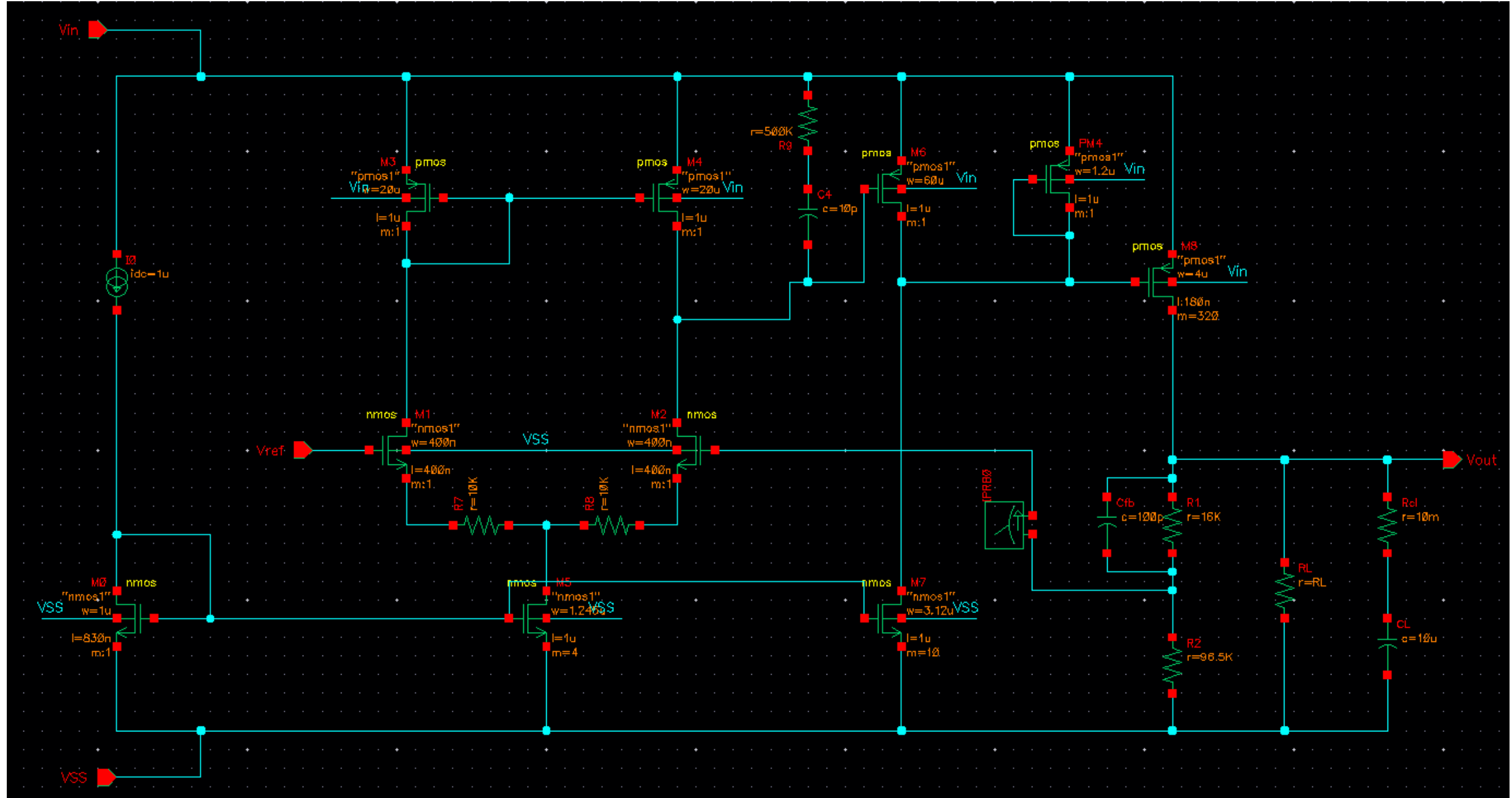
# CMOS Low Dropout (LDO) Voltage Regulator

By: Vansh Jain

# Specifications

| Parameter            | Specification        |
|----------------------|----------------------|
| Input Voltage Range  | 1.62V-1.98V          |
| Output Voltage       | 1.4 V (fixed)        |
| Maximum Load Current | 1 $\mu$ A – 50 mA    |
| Line Regulation      | $\leq 5$ mV/V        |
| Load Regulation      | $\leq 10$ mV         |
| Quiescent Current    | $\leq 50$ $\mu$ A    |
| PSRR                 | $\geq 40$ dB @ 1 kHz |
| Phase Margin         | $\geq 45^\circ$      |
| Output Capacitor     | 1 $\mu$ F (external) |

# Schematic



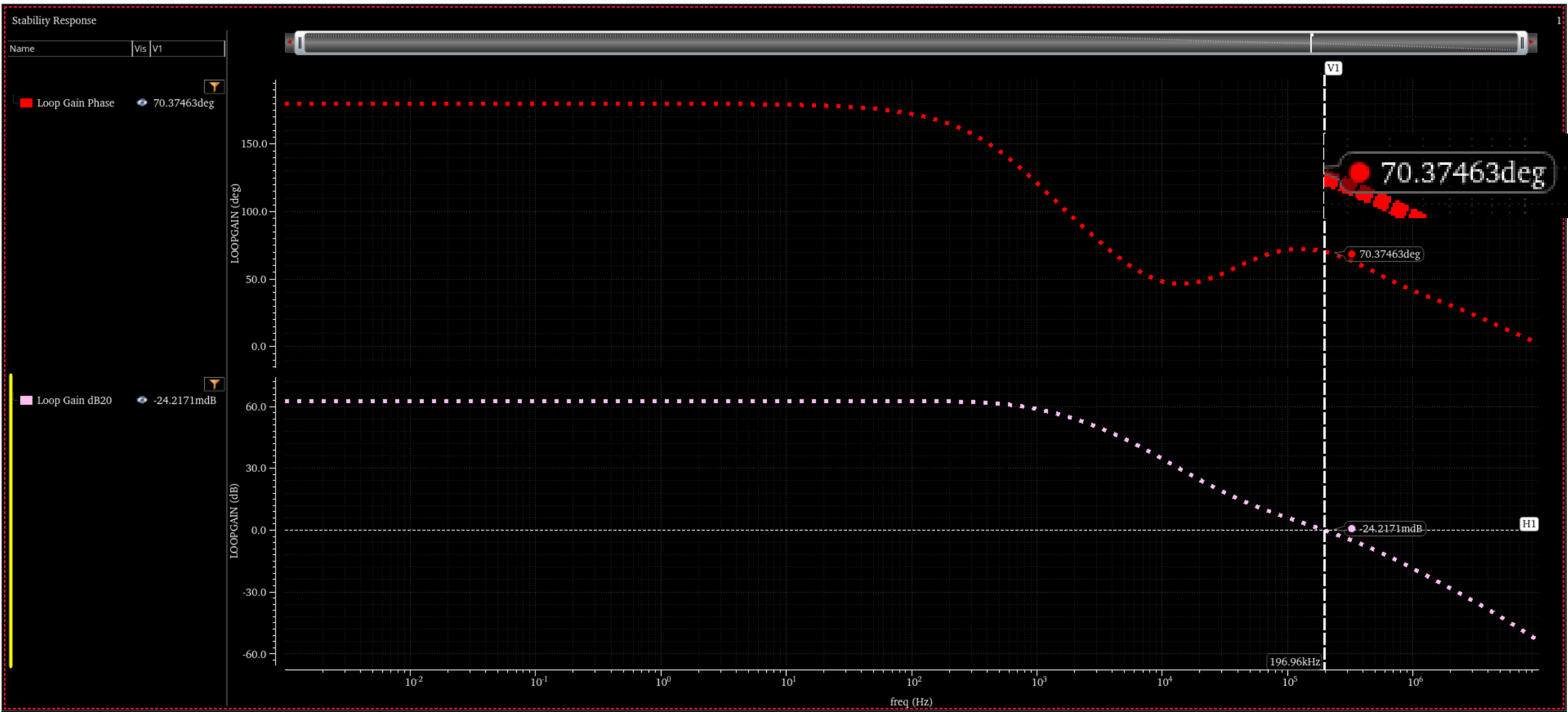
# Stability Analysis

4 Corner cases:

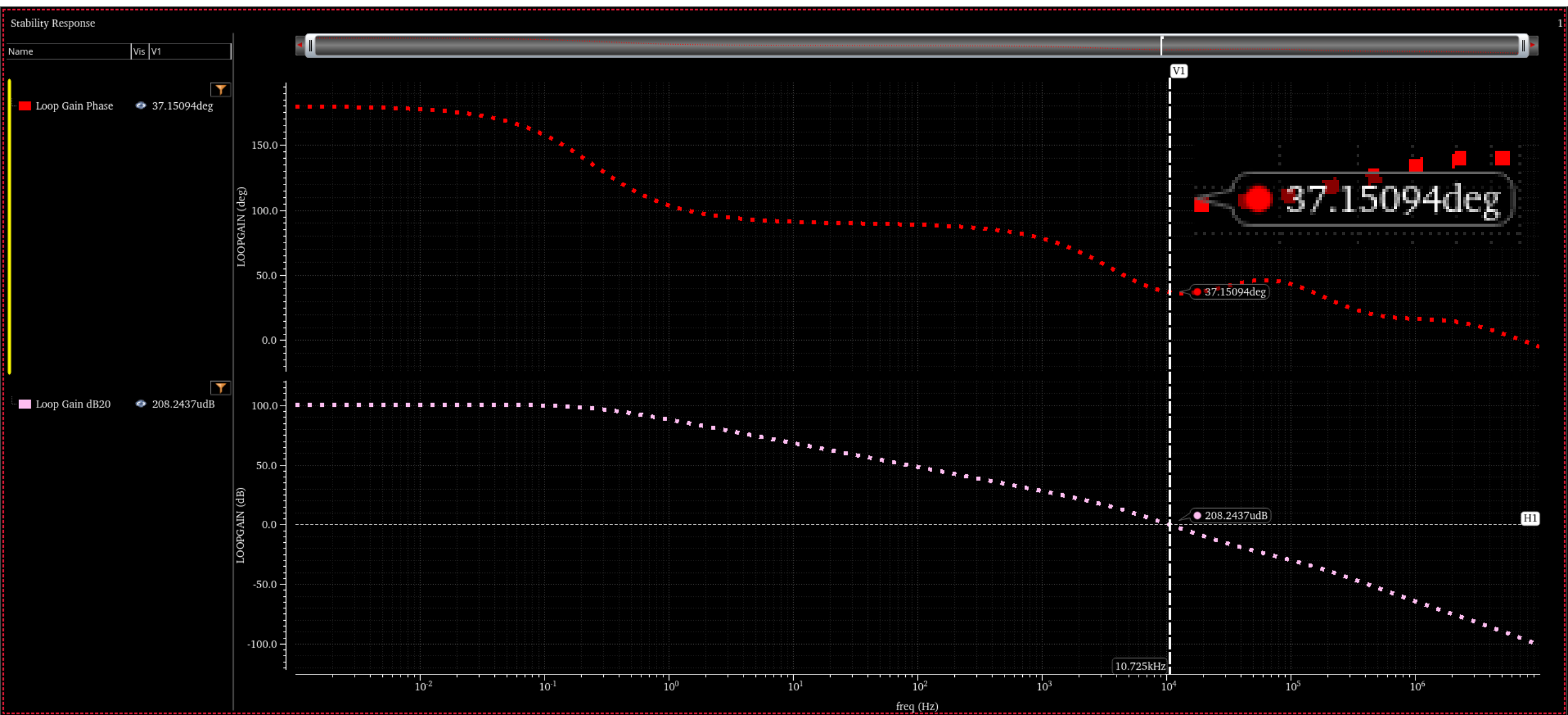
- $V_{in} = 1.98V$ , Load Current = 50mA
- $V_{in} = 1.98V$ , Load Current = 1uA
- $V_{in} = 1.62V$ , Load Current = 50mA
- $V_{in} = 1.62V$ , Load Current = 1uA

# Case 1: $V_{in} = 1.98V$ , Load Current = 50mA

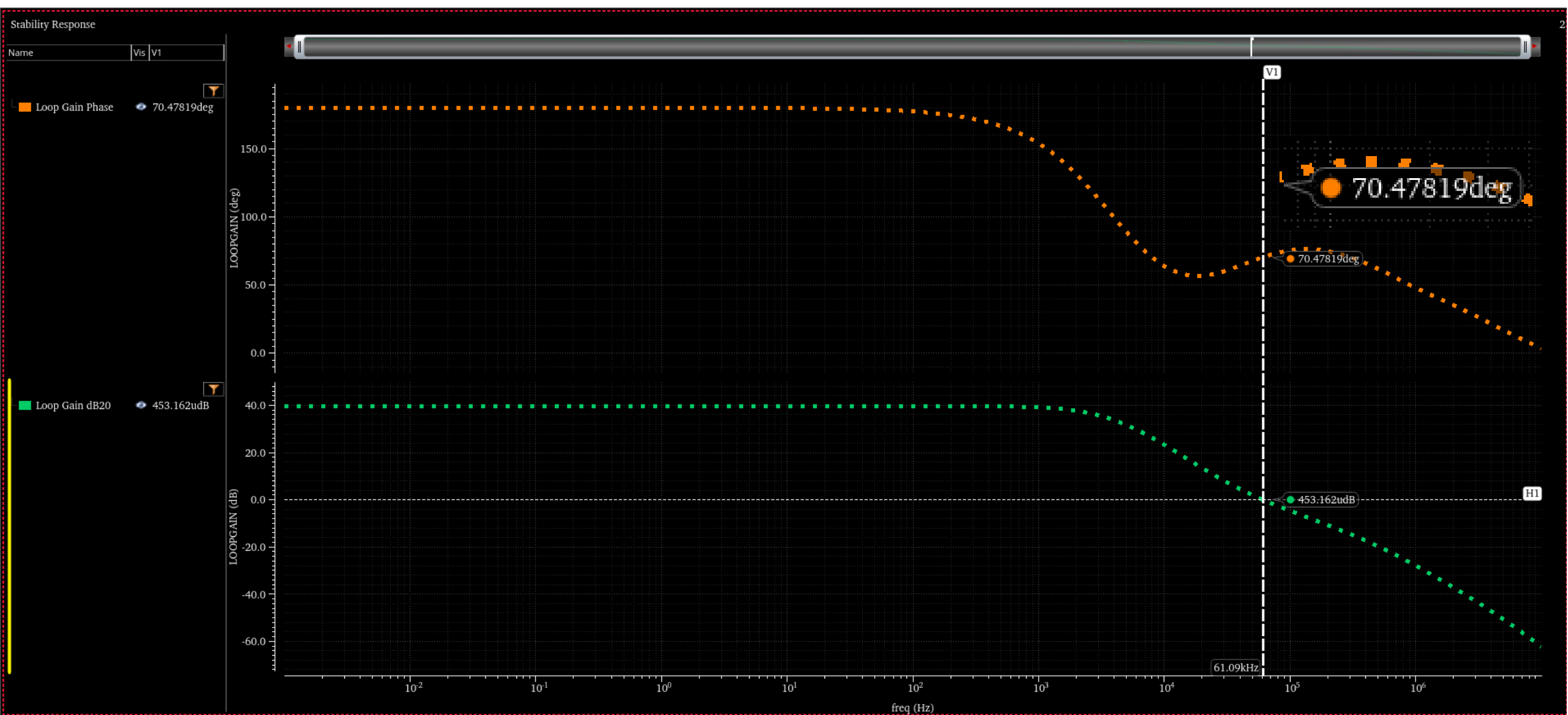
{Note: Even the lowest dip here before the UGB is 46 degrees.}



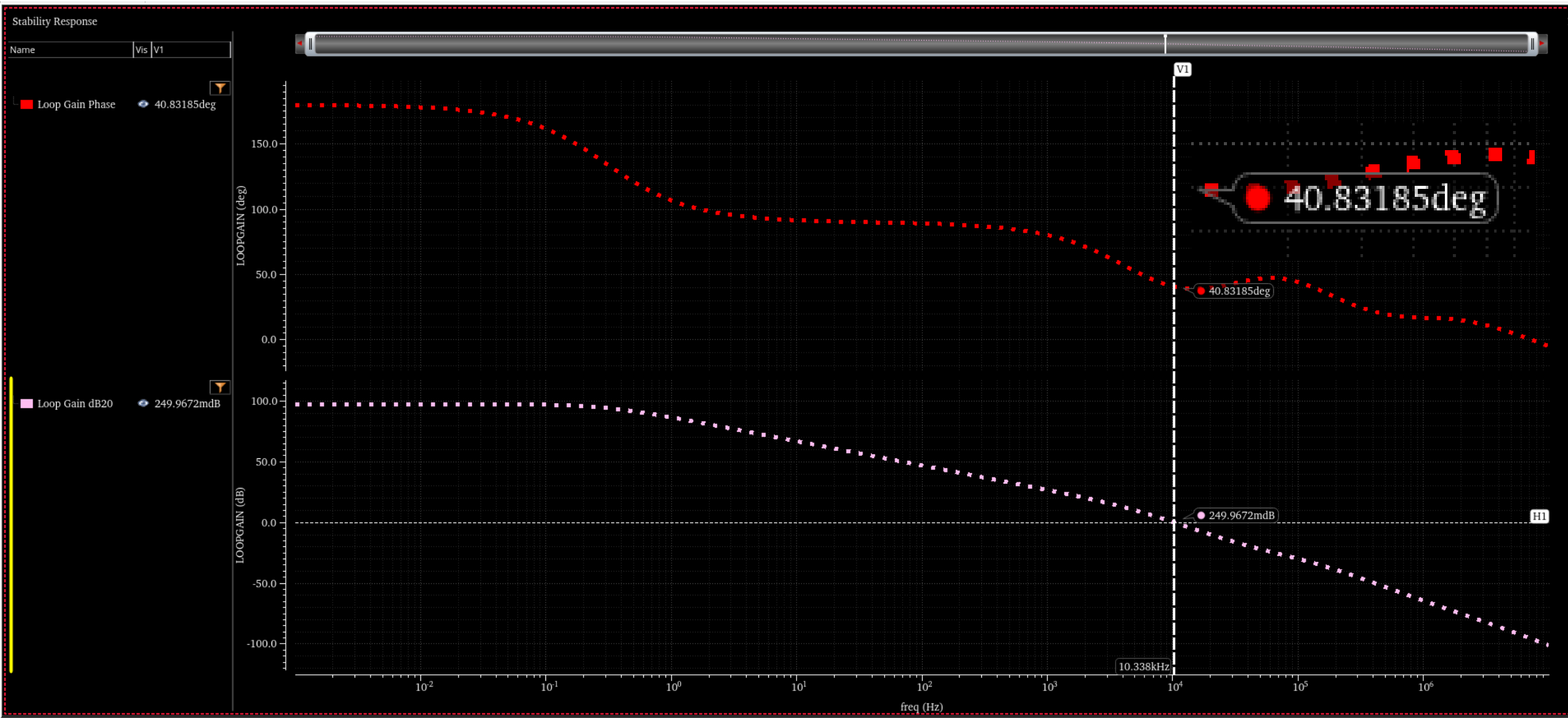
# Case 2: $V_{in} = 1.98V$ , Load Current = $1\mu A$



# Case 3: $V_{in} = 1.62V$ , Load Current = 50mA



# Case 4: $V_{in} = 1.62V$ , Load Current = $1\mu A$



# Load Regulation

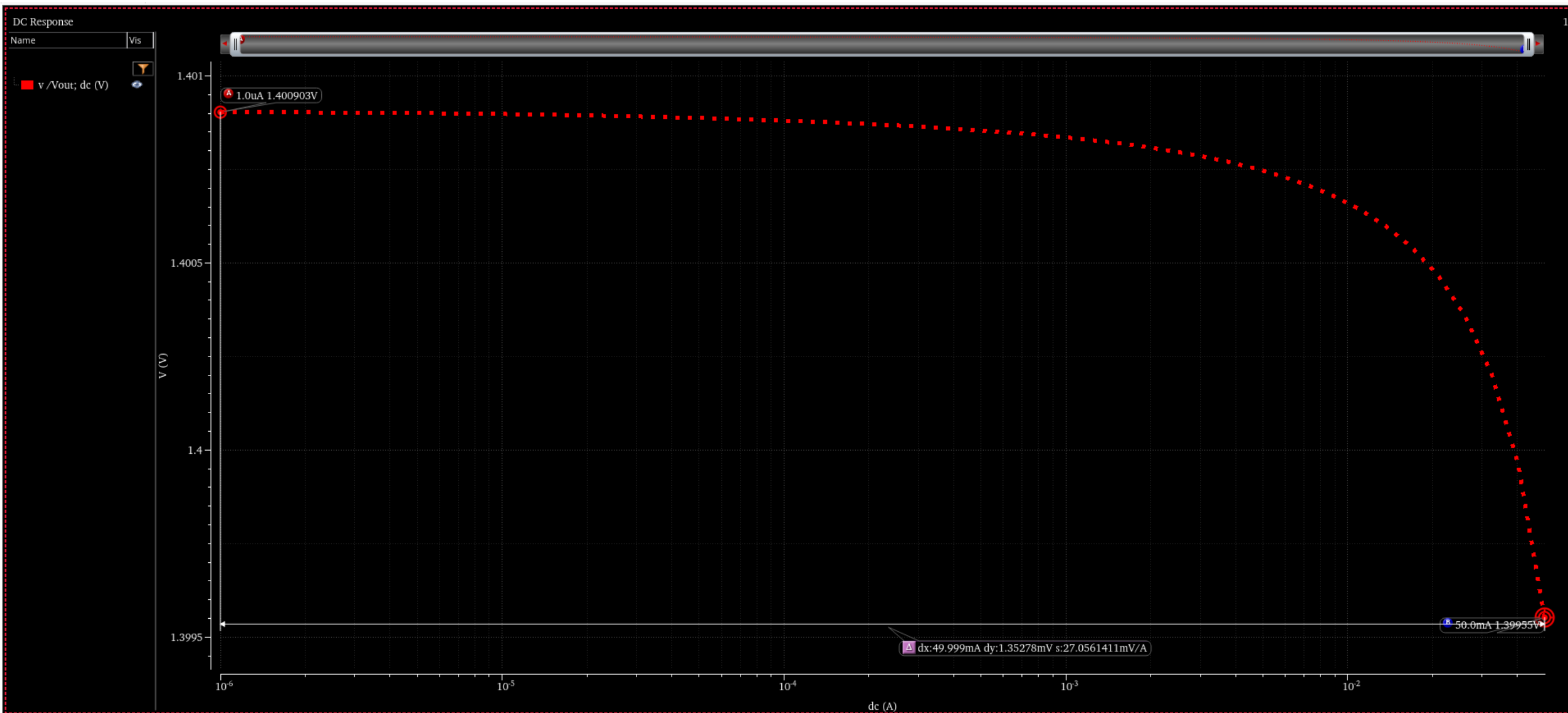
2cases:

- $V_{in} = 1.98V$
- $V_{in} = 1.62V$

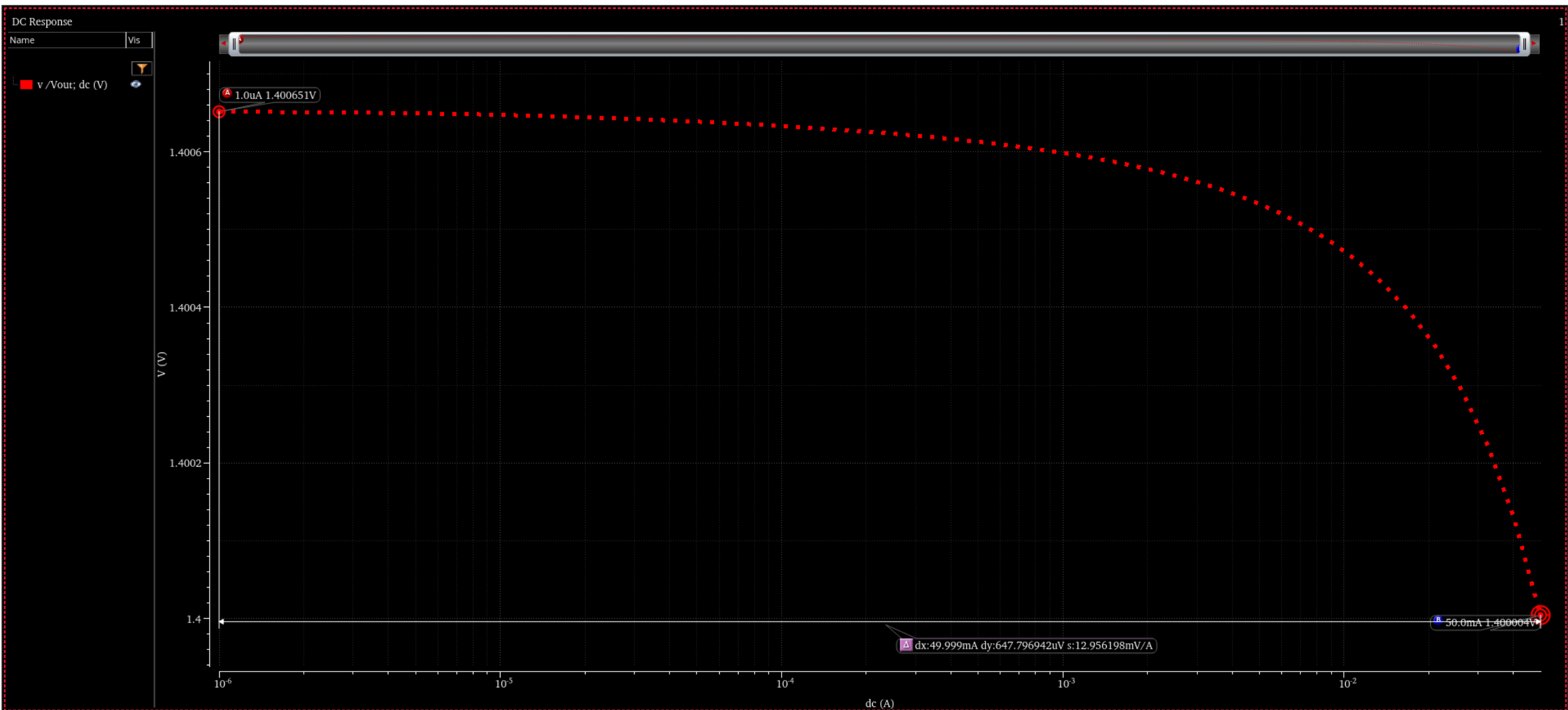
Formula Used:

- Load Regulation =  $\frac{\Delta V_{out}}{\Delta I_{load}}$

At Vin = 1.62V, Load Regulation = 27mV/A



At  $V_{in} = 1.98V$ , Load Regulation =  $12.95mV/A$



# Line Regulation

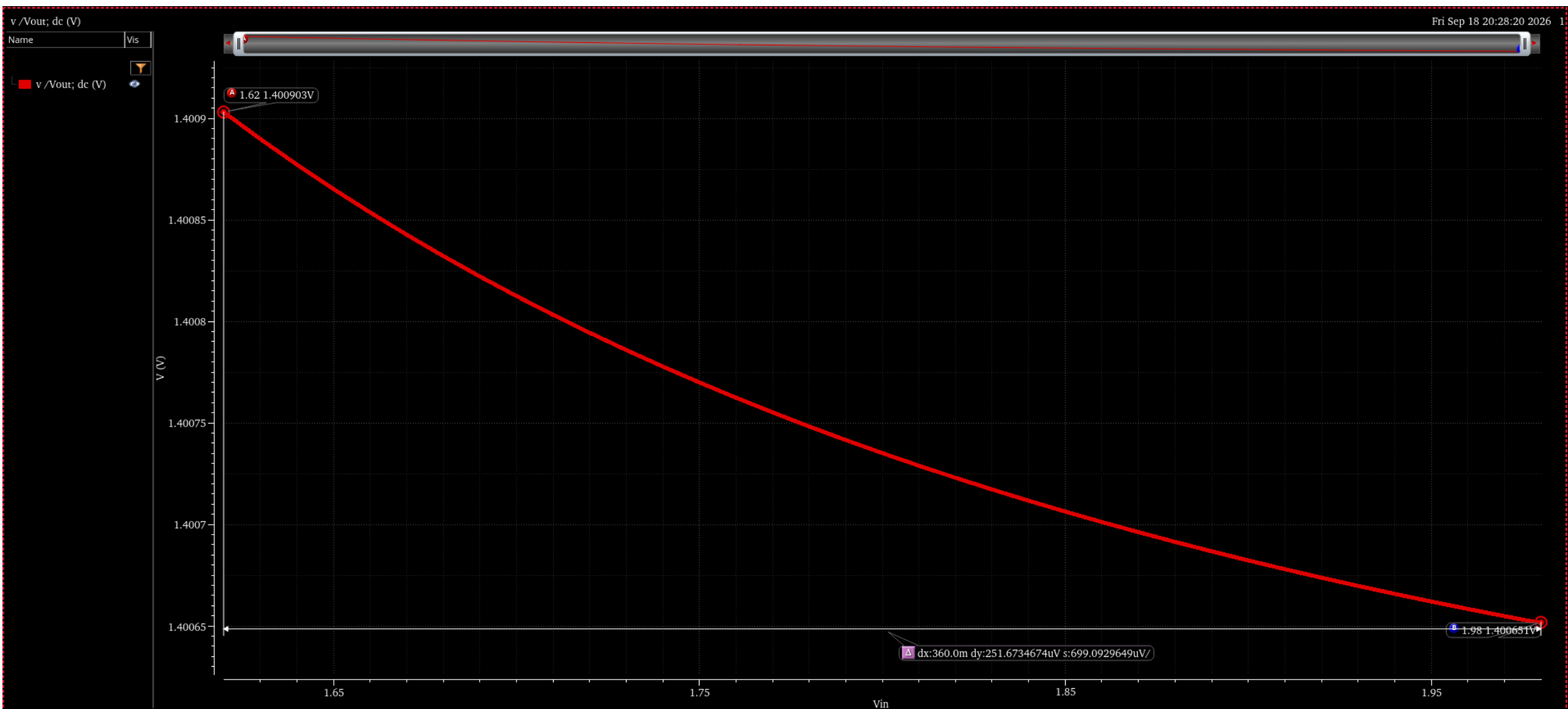
2cases:

- Load Current = 1uA
- Load Current = 50mA

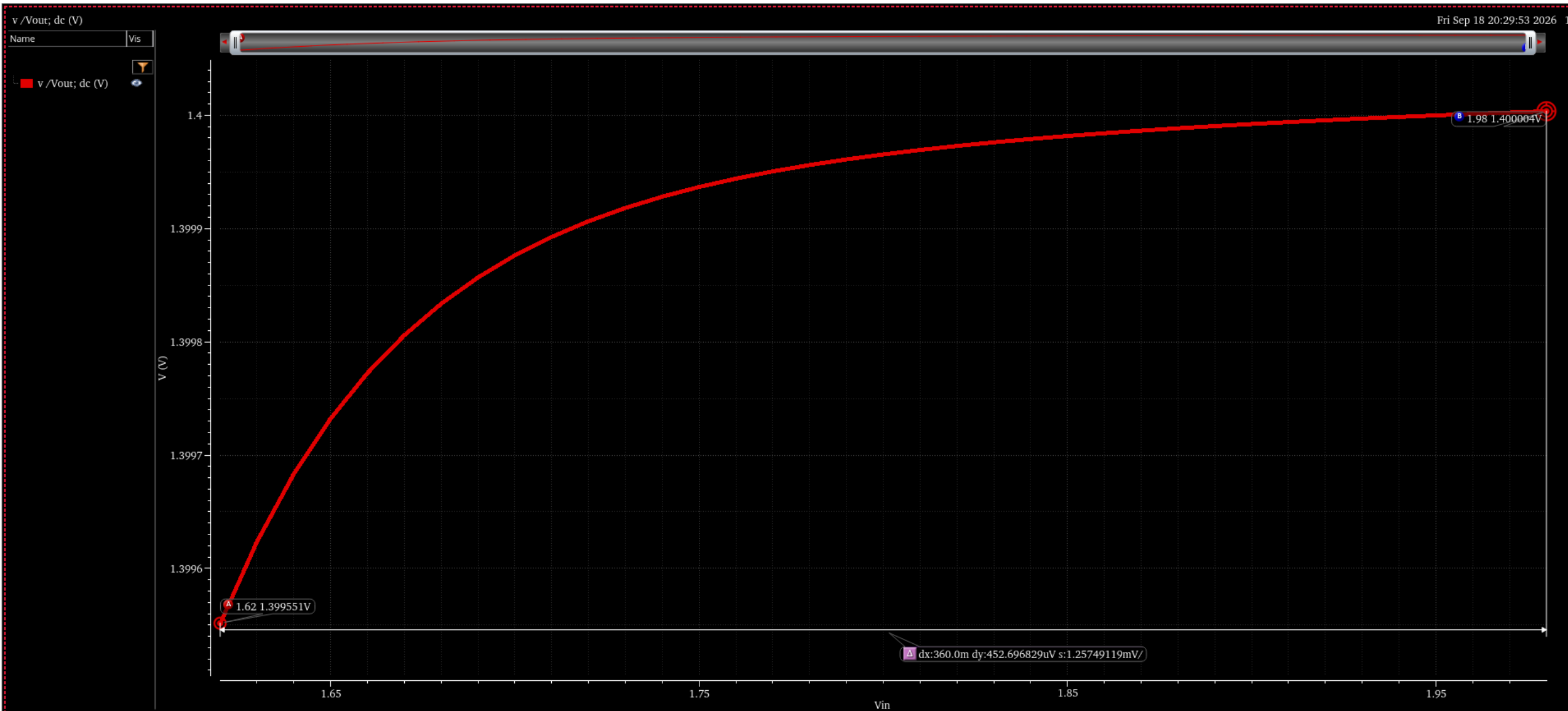
Formula Used:

- Line Regulation =  $\frac{\Delta V_{out}}{\Delta V_{in}}$

# At 1uA, Line regulation = 699.09 $\mu\text{V}/\text{V}$



At 50mA, Line regulation = 1.257mV/V



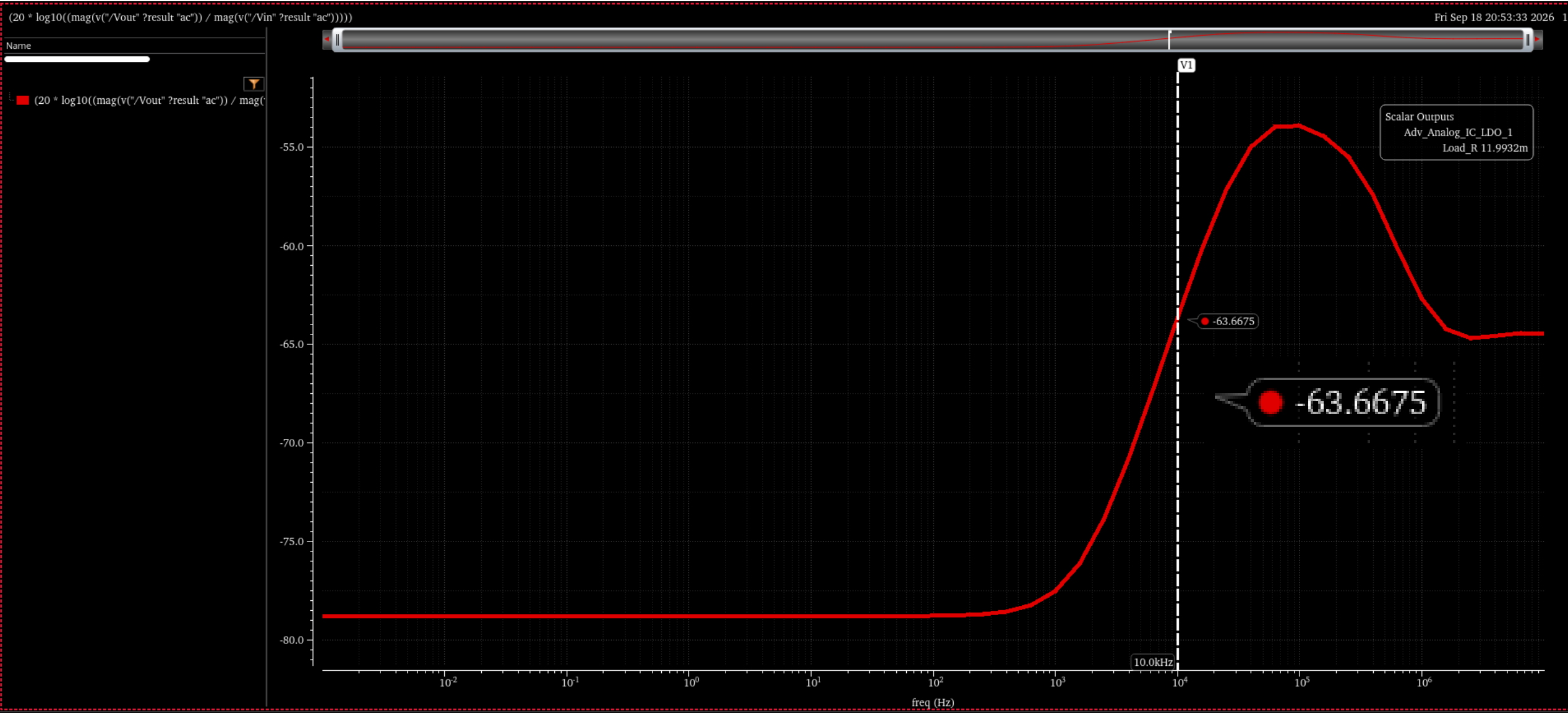
# PSRR

4 Corner cases:

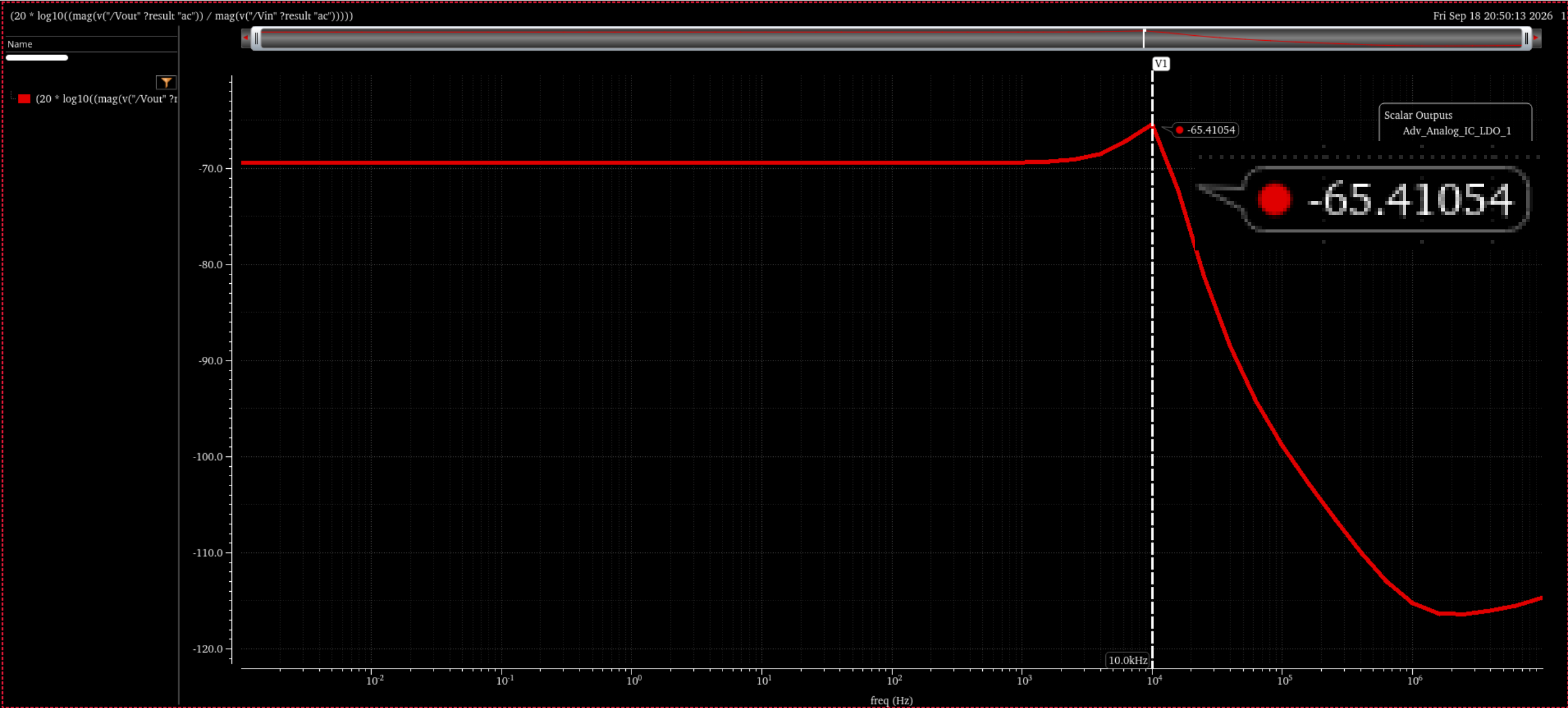
- $V_{in} = 1.98V$ , Load Current = 50mA
- $V_{in} = 1.98V$ , Load Current = 1uA
- $V_{in} = 1.62V$ , Load Current = 50mA (The worst scenario)
- $V_{in} = 1.62V$ , Load Current = 1uA

Formula:  $PSRR = 20 * \log\left(\frac{V_{in}}{V_{out}}\right)$

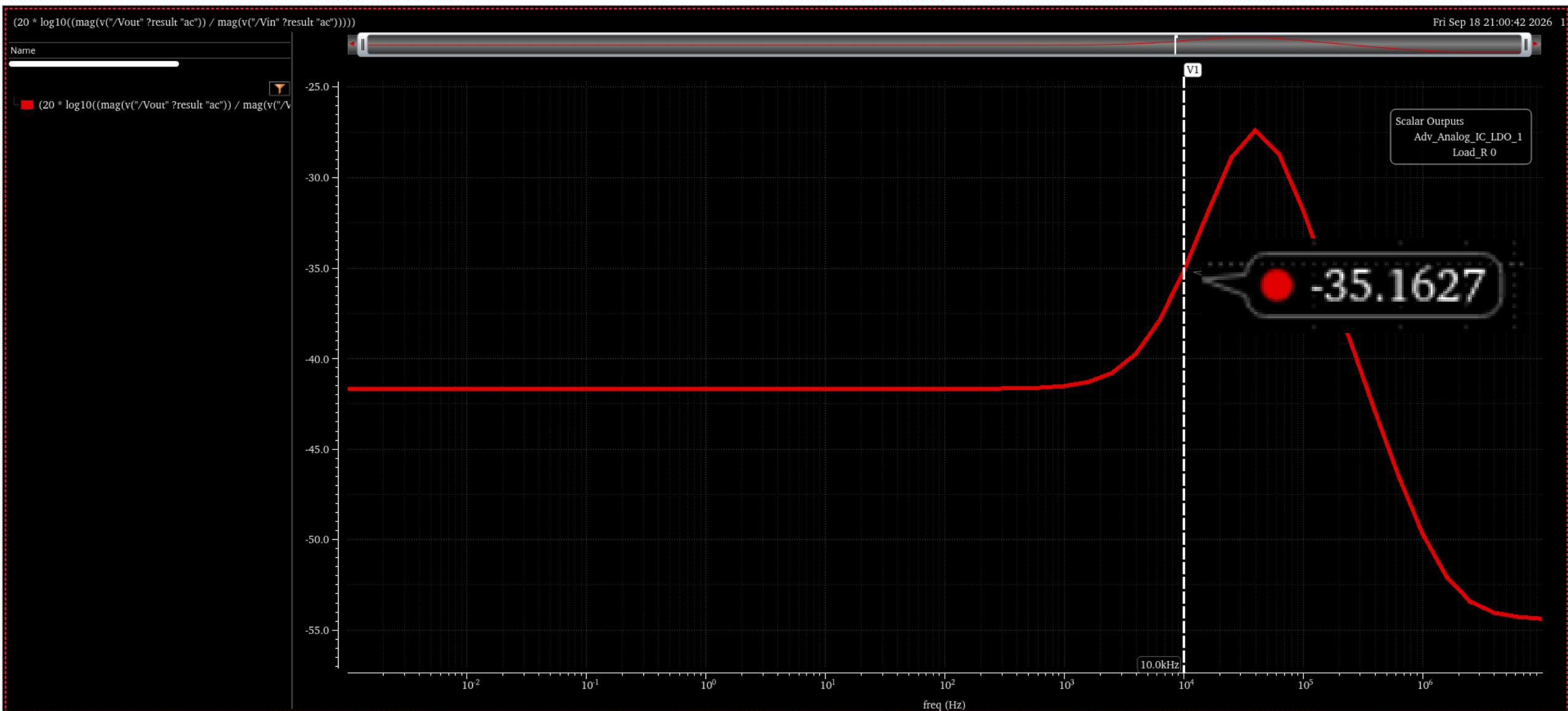
# Case 1: $V_{in} = 1.98V$ , Load Current = 50mA



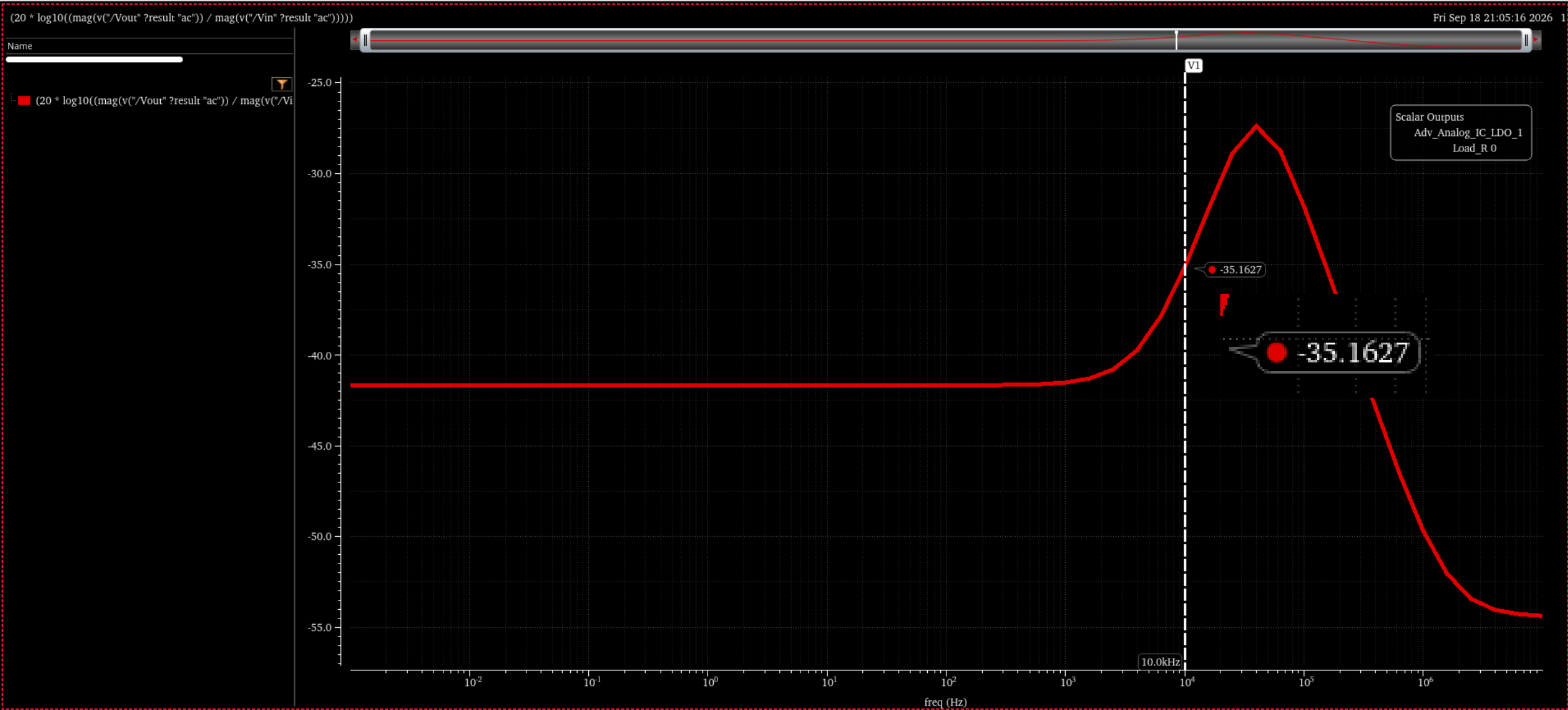
# Case 2: Vin = 1.98V, Load Current = 1uA



# Case 3: Vin = 1.62V, Load Current = 50mA



# Case 4: $V_{in} = 1.62V$ , Load Current = $1\mu A$

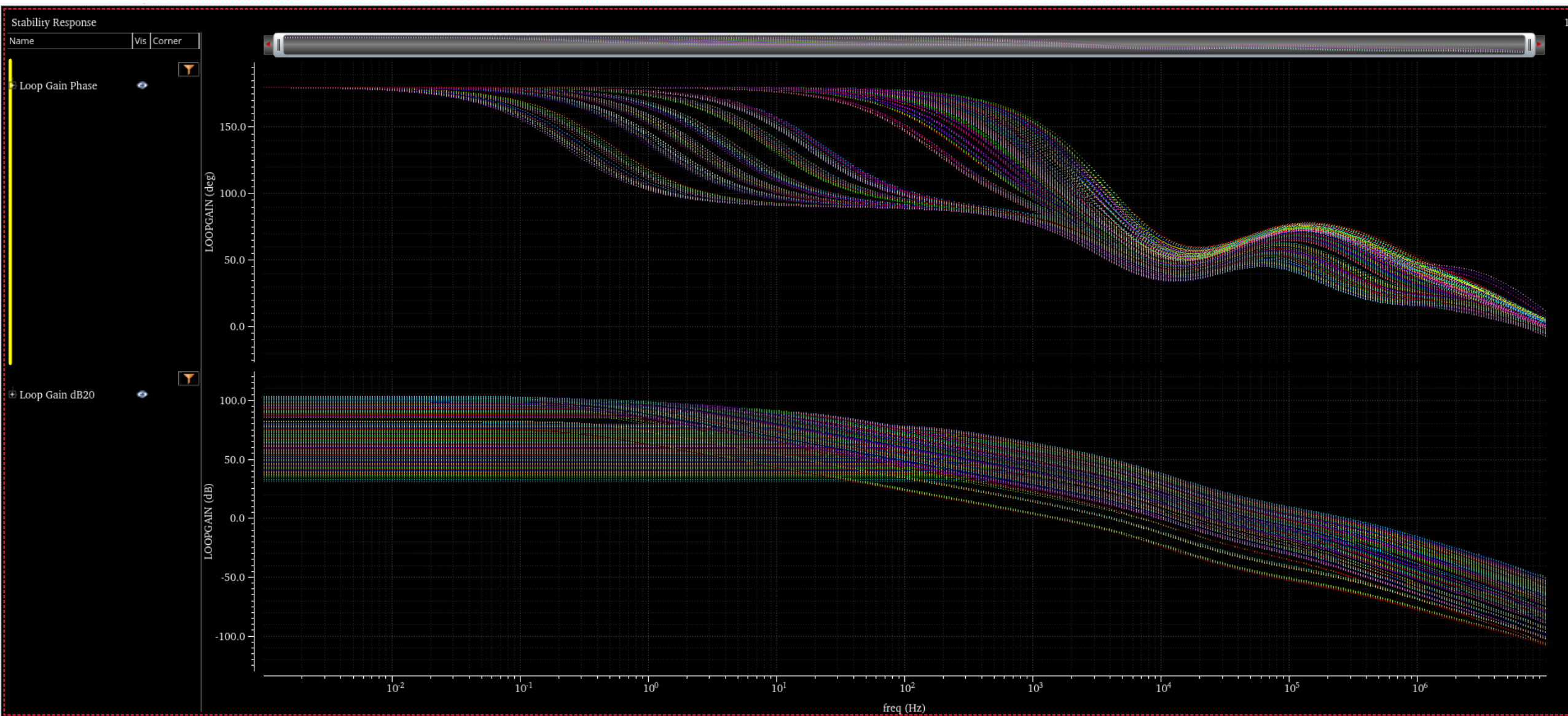


# Corner Testing

Conditions set for corner testing:

- Corners: SS, SF, FS, FF
- Temperature = -40°C, 30°C, 125°C
- Vin = From 1.62, 1.8V, 1.98V
- Load Current = 1uA, 10uA, 50uA, 100uA, 200uA, 500uA, 1mA, 10mA, 20mA, 30mA, 35mA, 40mA, 45mA, 50mA
- Total Number of corners = 505

# Gain and Phase Margin

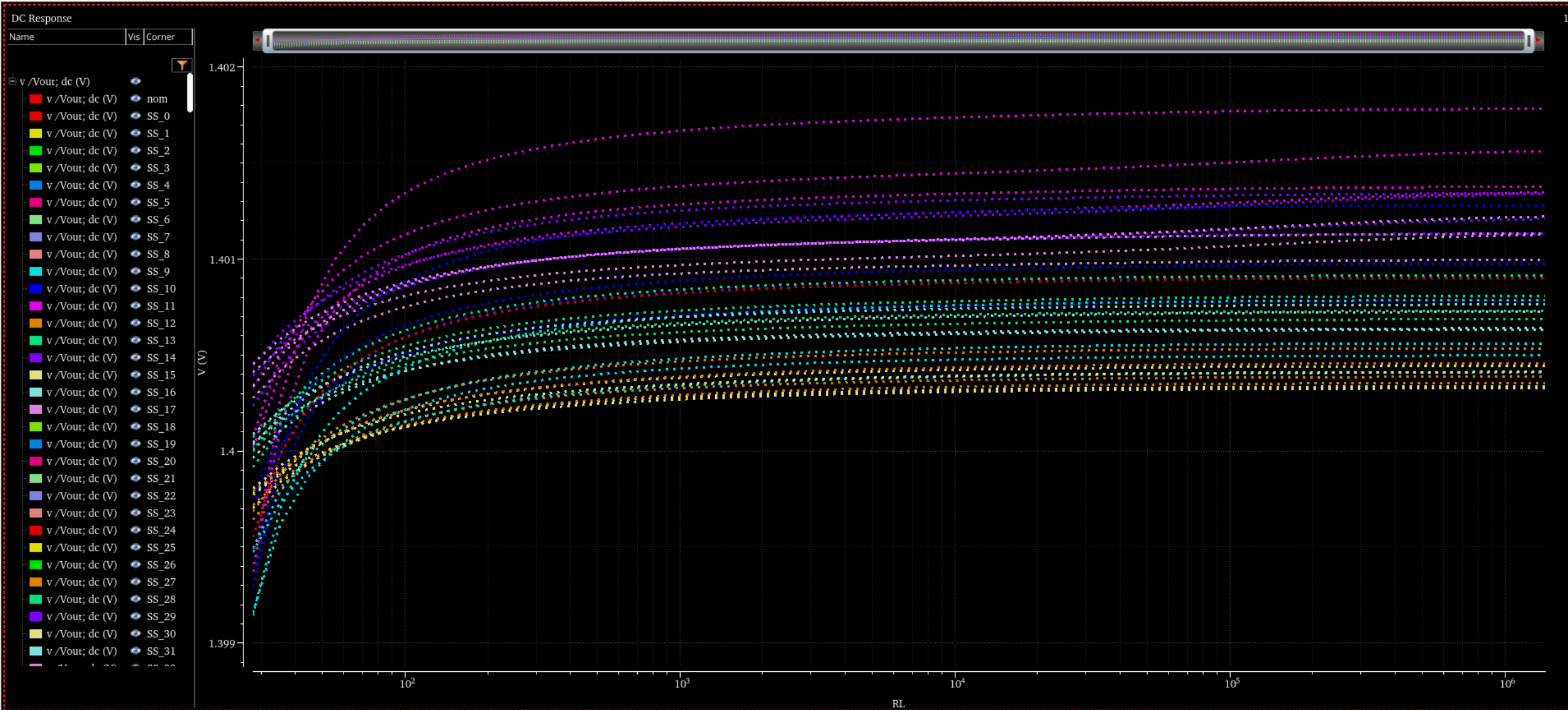


# Observation of Stability Plot

Phase Margin Insights: (Specification Requirement:  $\leq 45$  deg)

- Highest: 77.62 deg
- Lowest: 35 deg
- Insight: PM is incredibly stable and above 45 deg for load currents above 100uA. Below that it becomes a little underdamped in some corners.

# Load Regulation

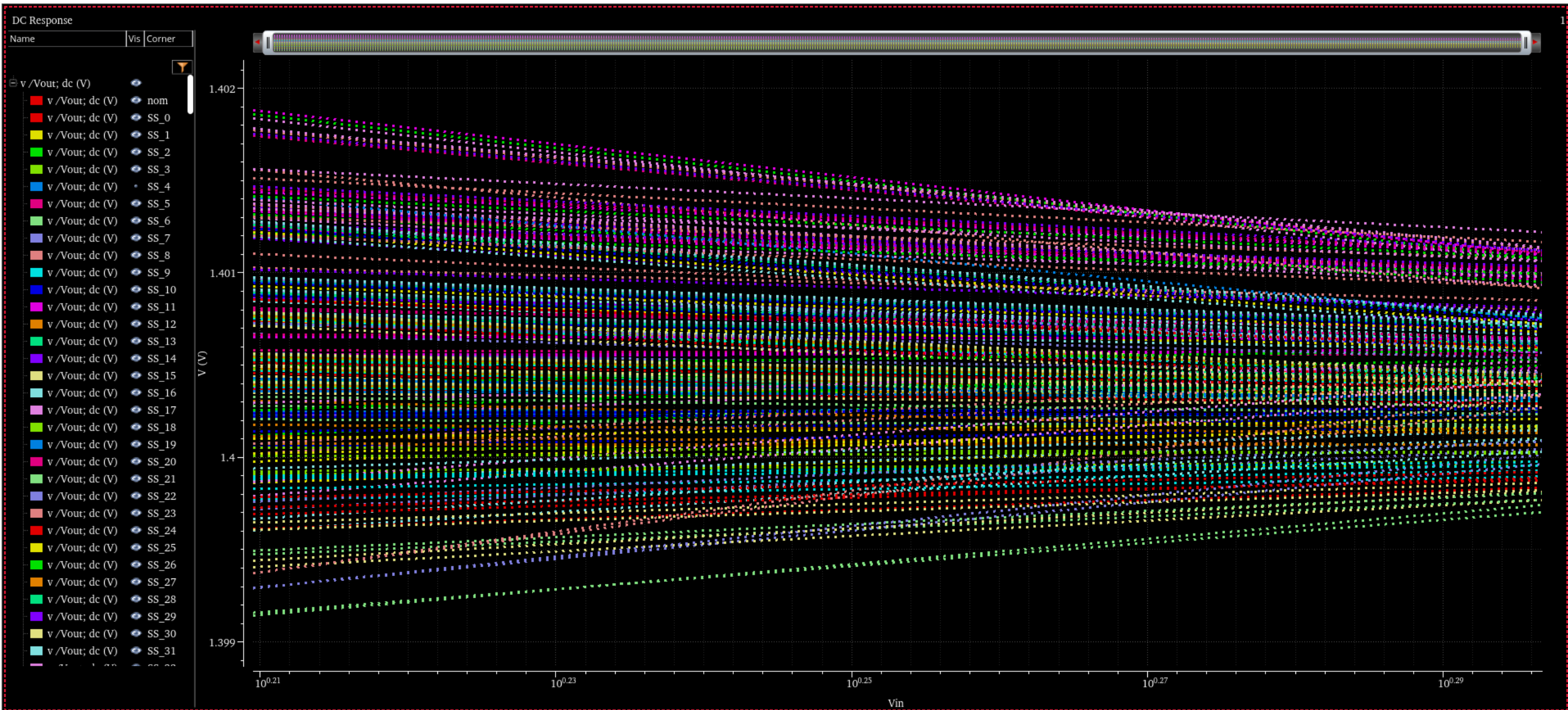


# Observation of Load Regulation Plot

Load Regulation Insights: (Specification Requirement:  $\leq 10\text{mV}$ )

- Best (lowest) Droop:  $13\text{mV}$  ( $0.66\text{mV}$ )
- Worst (highest) Droop:  $48\text{ mV}$  ( $2.41\text{mV}$ )
- Insight: The output is exceptionally stiff and consistent. Load Regulation degrades very slightly (voltage drops a bit more) as the silicon temperature heats up to  $125\text{ deg}$ , but remains highly accurate across the entire  $1\text{mA}$  to  $50\text{mA}$  sweep.

# Line Regulation

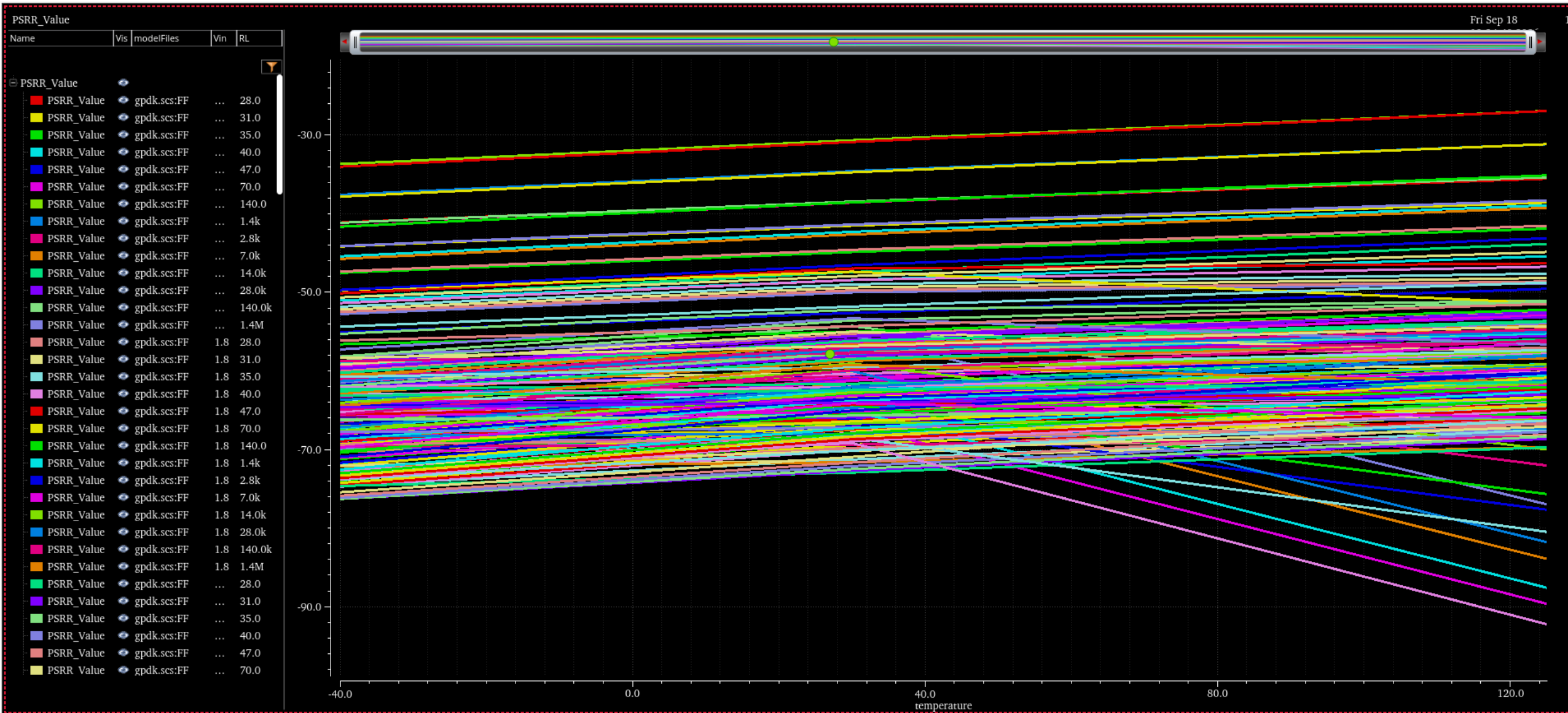


# Observation of Line Regulation Plot

Line Regulation Insights: (Specification Requirement:  $\leq 5\text{mV/V}$ )

- Lowest:  $-2.16\text{mV/V}$
- Highest:  $+2.68\text{ mV/V}$
- Insight: The feedback loop successfully rejects DC Battery fluctuations, keeping the output shift tightly bounded within a  $\pm 3\text{mV}$  window. Variance spreads out slightly at high temperatures ( $125\text{ deg Celcius}$ ), but the absolute shift remains negligible.

# PSRR



# Observation of PSRR Plot

PSRR Insights: (Specification Requirement:  $\leq 40\text{dB}$  @ 10kHz)

- Best (Highest): 92.24 dB
- Worst (Lowest): 26.93 dB
- Insight: Out of all the 500 corners, the PSRR lies below 40dB for exactly 24 cases. These cases generally include  $V_{in}$  of 1.62V and Heavy load current above 35mA.